

Toni Zeng

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Education

University of Waterloo

Expected Apr 2027

Bachelor of Applied Science in Mechatronics Engineering (CO-OP) | **CGPA: 3.8/4.0**

Relevant Courses: Data Structures & Algorithms, Real-Time Operating Systems, Digital Computation

Work Experience

Shopify | **Infrastructure Engineering Intern, MySQL Infrastructure**

January 2026 – August 2026

- Saved **\$800K annually** by reclaiming **300 TB** of **MySQL** storage, creating a metrics job to identify fragmented tables, and coordinating cross-team **InnoDB table compaction** and disk shrinking.
- Built the knowledge foundation for a company-wide **AI incident-response agent** by documenting **ProxySQL architecture**, scaling strategies, and incident playbooks as reusable **Claude skills**, reducing resolution time by **10%**.
- Reduced migration risk from **MySQL** to **YugabyteDB** by removing **40+** deprecated schema components and updating production schemas for compatibility.
- Validated prototype **ProxySQL** architectures under production-like traffic by designing load tests on canary shards, identifying performance bottlenecks, and comparing system behaviour.
- Executed production **MySQL writer failovers** during scheduled maintenance, coordinating traffic migration and validating database health to prevent service disruption.

Shopify | **Software Engineering Intern, Database Services**

May 2025 – August 2025

- Built an internal **MCP server** in **Ruby on Rails** for schema migration validation, reducing production rollbacks by **10+ per month** and supporting **200+ engineers**.
- Led **zero-downtime deployments** of **30+ ZooKeeper instances** on **Kubernetes** in **GCP** using **Terraform**.
- Reduced **p95 query latency** by **18%** by identifying overloaded database partitions and redistributing high-traffic tenant workloads to reduce resource contention.
- Reduced manual scaling and restarts by **85%** by implementing autoscaling and recovery logic for **Kubernetes** workers that execute shop migrations.

Kinaxis | **Software Developer Intern**

September 2024 – December 2024

- Reduced deployment errors by **20%** by refining **Dockerized applications** and **Helm charts** through standardized image builds, resource limits, and logging conventions, improving overall system consistency.
- Delivered **15** production fixes to **C# .NET** services, refactoring legacy modules and modernizing outdated unit tests to improve test coverage, performance, and maintainability.
- Optimized **Datadog** log pipelines to cut processing latency by **25%** and improve alert reliability across **50+** services.

Royal Bank of Canada | **Software Engineering Intern**

January 2024 – April 2024

- Reduced code inconsistencies by **91%** by developing production validation **CI/CD pipelines** in **Java** and **Jenkins** to compare legacy code with **GitHub** sources and flag mismatched files with detailed error reports.
- Decreased deployment time by **35%** by streamlining **Docker** image builds and parallelizing **Jenkins** stages.
- Deployed **DevOps infrastructure** with **UNIX** on **30+** Mac Minis, enabling department-wide mobile deployments.
- Implemented a **React-based** developer portal for **90+** employees, centralizing resources and improving accessibility.

Metrolinx | **Data Engineering Intern**

May 2023 – August 2023

- Improved **ETL efficiency** by **98%** by implementing automated file validation using **Python** scripts.
- Developed optimized **SQL queries** to extract and transform data for **10+** customer-facing **Power BI** dashboards.

Skills

Languages: Python, Java, Ruby, C++, C#, SQL, JavaScript

Data & Observability: MySQL, InnoDB, ProxySQL, Prometheus, Grafana, Datadog

Infrastructure & Tools: Kubernetes, Terraform, Docker, GCP, Apache ZooKeeper, Jenkins, Git